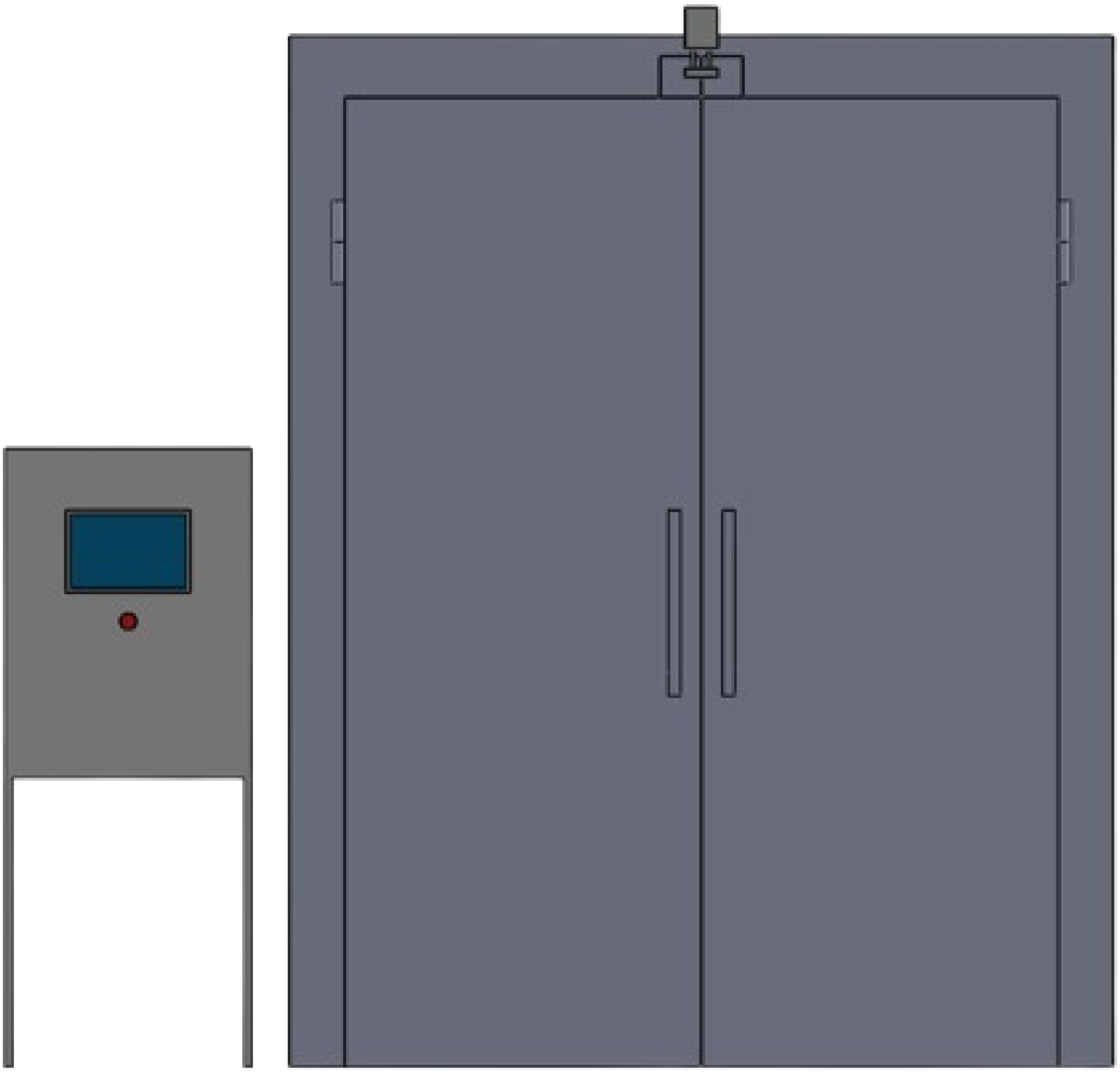
MANUAL

UV TUNNEL



CONTENT

1. Introduction

Preface

This manual contains essential information regarding the equipment and the proper operation of the UV TUNNEL. It includes critical guidelines designed to prevent potential accidents and serious damage during operation. Users must familiarize themselves with the machine's controls and strictly adhere to the provided instructions. We strongly emphasize the importance of proper training for the safe handling of this machinery. This manual should always be kept accessible near the machine.

Note: This document has been carefully reviewed and filtered to remove confidential proprietary information. It is explicitly permitted for public distribution and inclusion in professional portfolios.

2. General Information

2.1 UV TUNNEL

The UV TUNNEL is situated between the Cooked Zone and the Raw Zone. The operation involves transporting food carts from the Raw side into the tunnel. While the carts are inside, UV lamps are utilized to sterilize and eliminate pathogens. Subsequently, the food is transferred to the Cooked Zone. The UV Tunnel is operated via control panels located on controller cabinets on both sides.

2.2 Safety Guidelines

1. Ensure that the doors are completely closed and secured before every operation.

2. Do not look directly at the UV lamps while they are active, as this may cause severe eye injury.

3. An Emergency button is located inside the UV Tunnel. This is specifically intended for use in the event of entrapment (being unable to exit the tunnel).

3. Technical Specifications

3.1 Preliminary Requirements

Machine Location

- The machine operates as the interface connecting the Raw Zone and the Cooked Zone.

- Sufficient clearance must be maintained around the machine to facilitate easy cleaning and maintenance.

Control Panel

- The controller cabinet must be installed in a dry location with appropriate environmental conditions and ambient temperature.

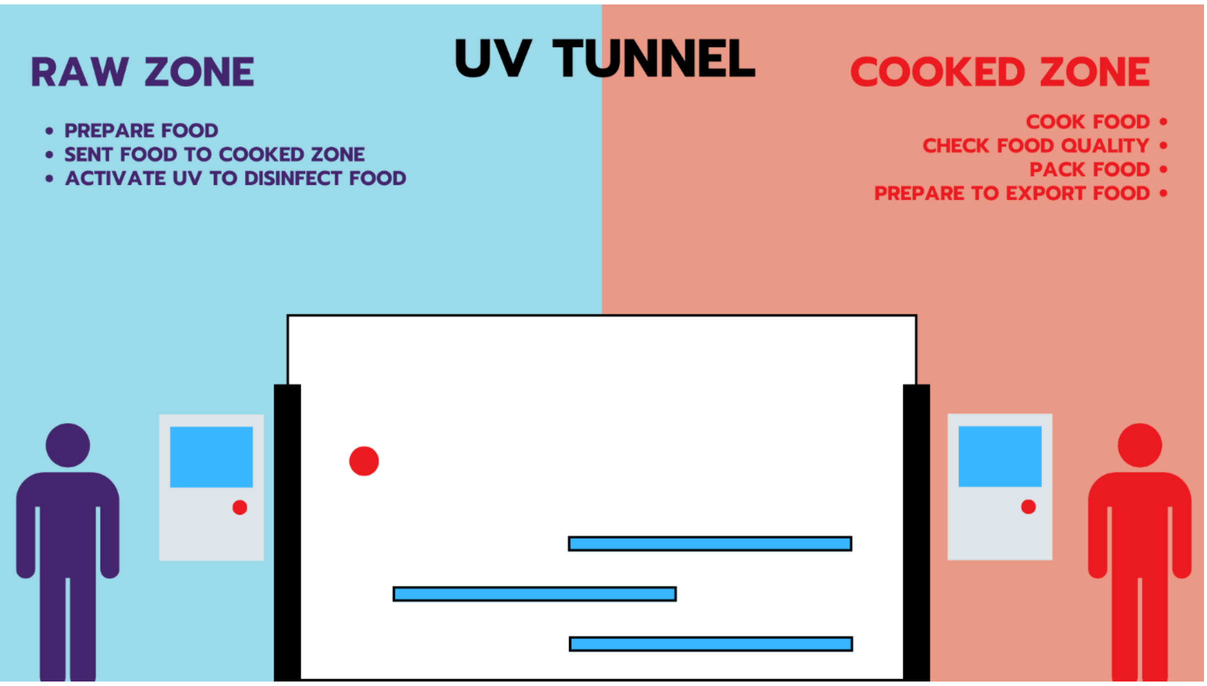
- The Raw Zone must provide ample space to ensure the control panel is easily accessible for operators.

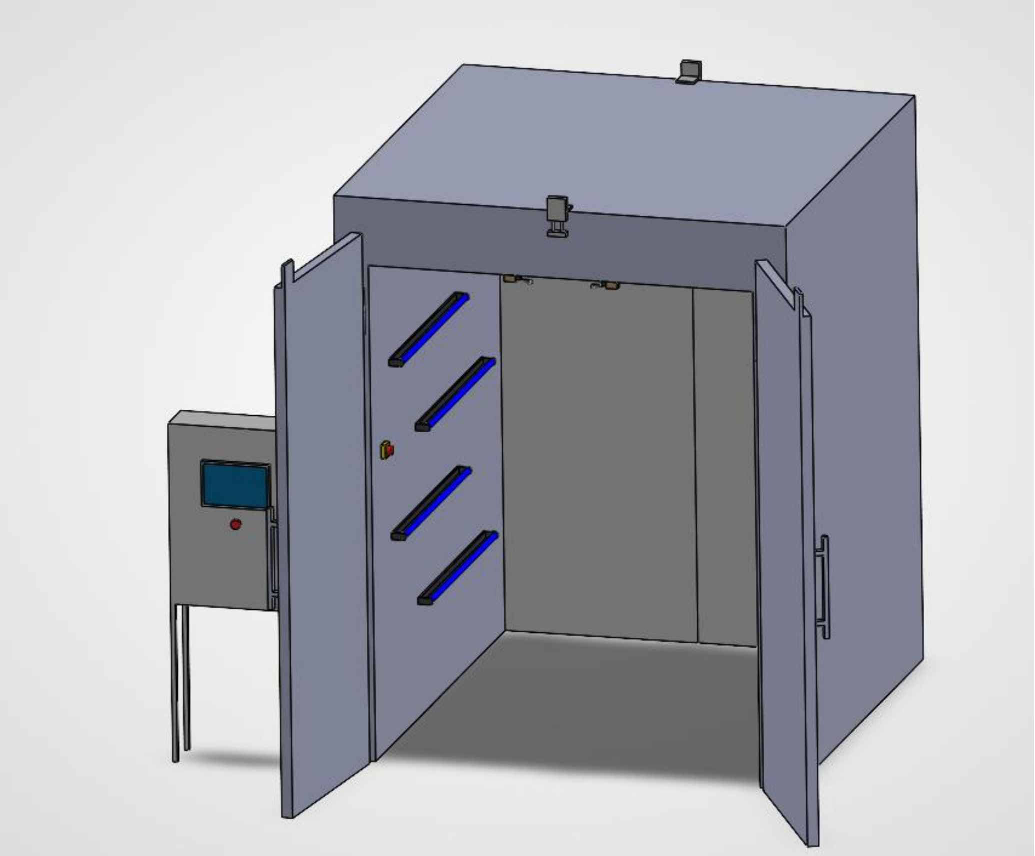
3.2 Power Requirements

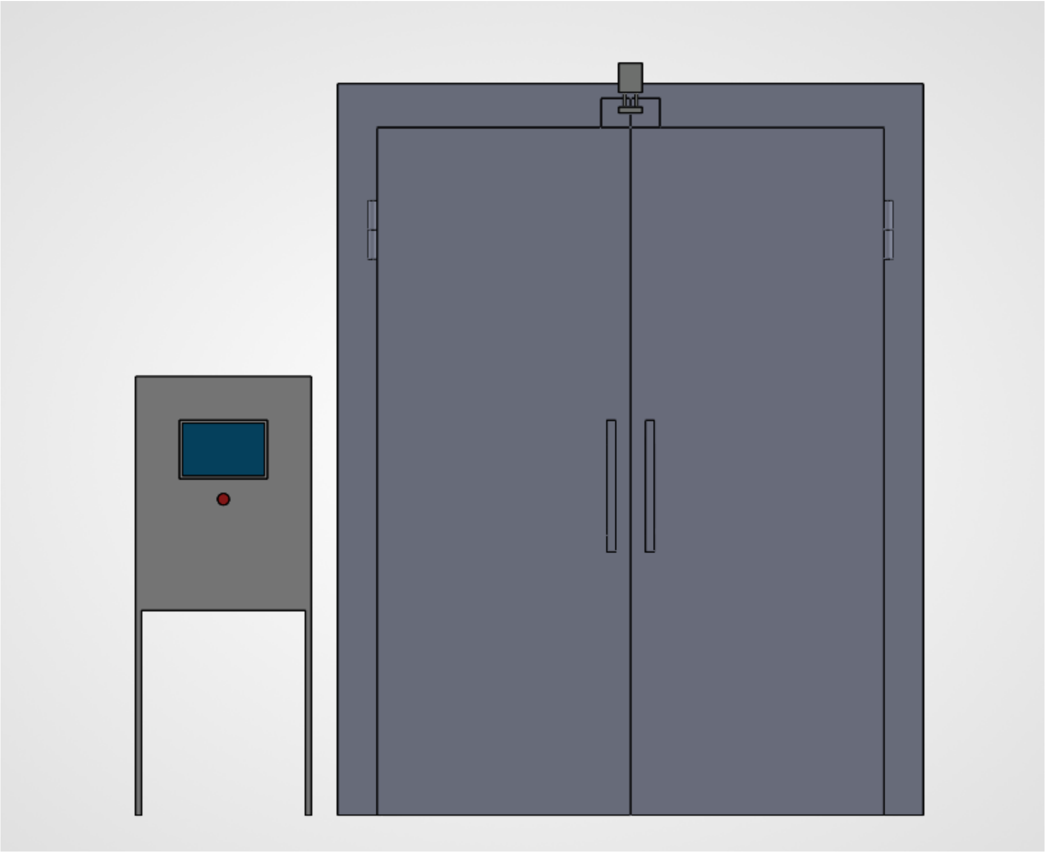
Power Supply: 380 VAC, 1-Phase, 50 Hz.

Total Current Consumption: 16 A.

4. UV TUNNEL View



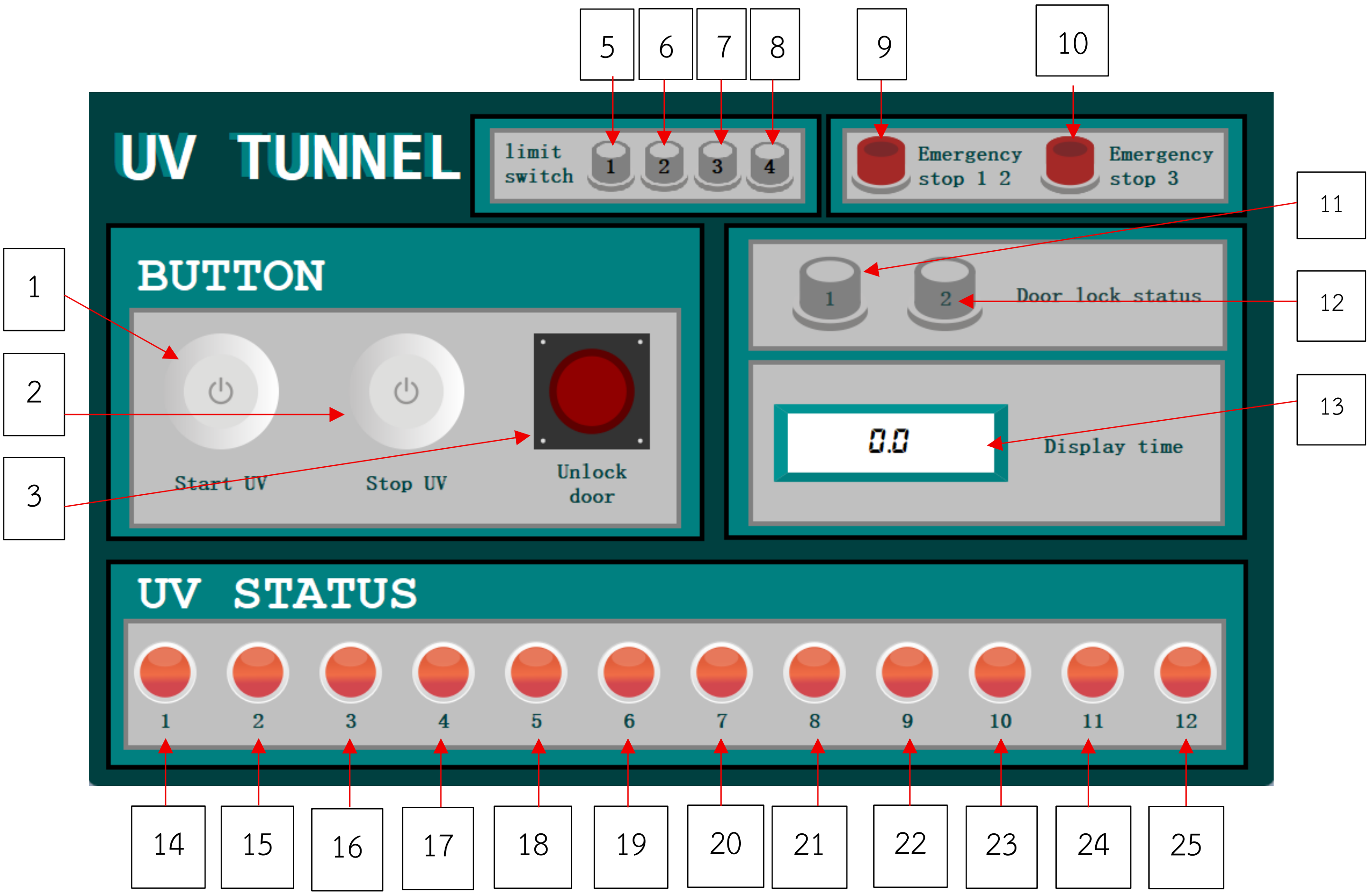




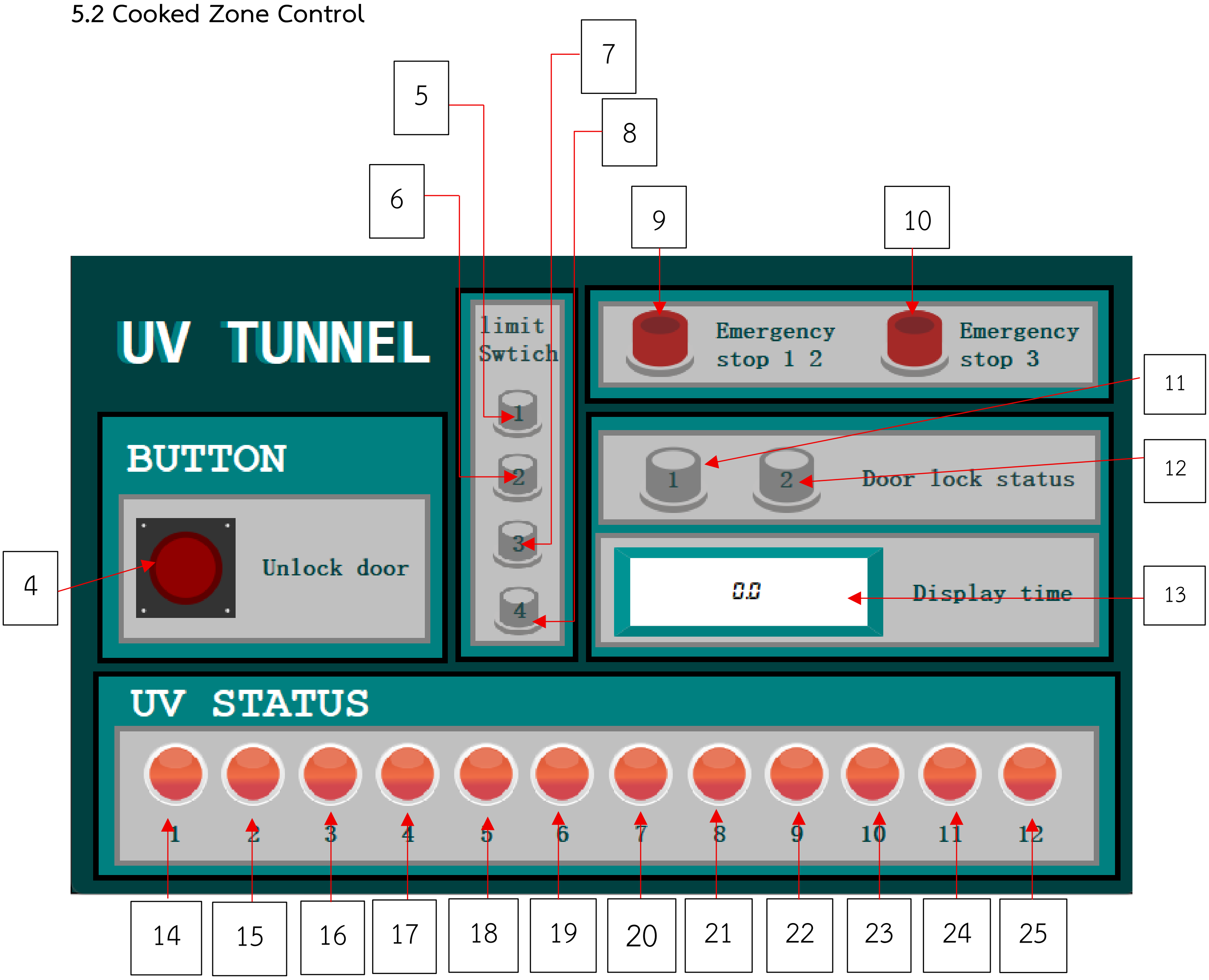
5. Operation

HMI

5.1 Raw Zone Control



5.2 Cooked Zone Control



5.3 HMI Button Descriptions

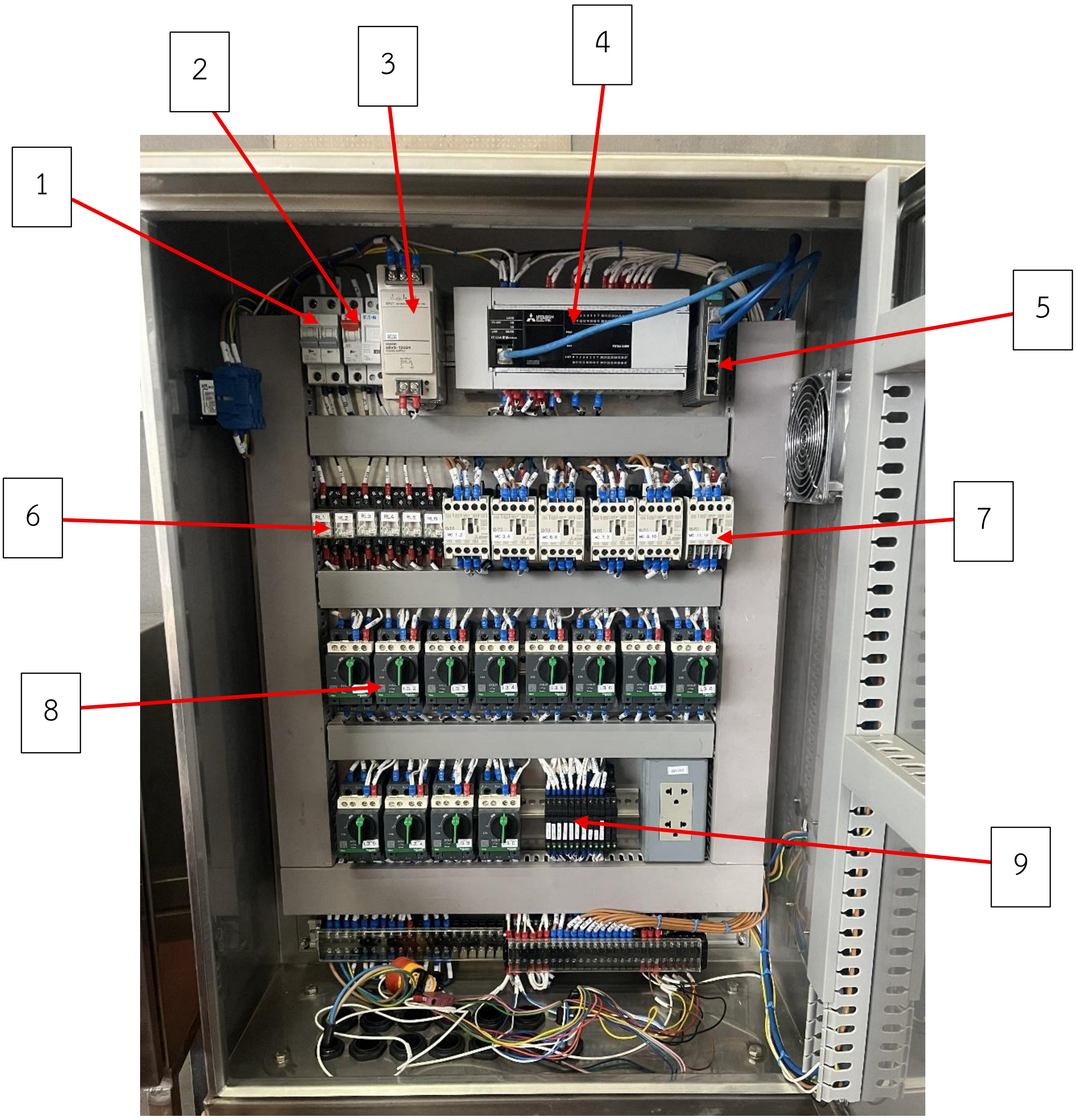
HMI control panel		
No.	Item	Operation
1	Start UV	Press to activate the UV lamps.
2	Stop UV	Press to deactivate all currently operating UV lamps.
3	Unlock door 1	- Unlocks the door on the Raw Zone side. - A flashing red indicator confirms that the door has been successfully unlocked.
4	Unlock door 2	- Unlocks the door on the Cooked Zone side. - A flashing red indicator confirms that the door has been successfully unlocked.
5, 6	Limit switch 1-2	- Displays the status of the Raw Zone limit switch. - Green: Switch is active / working. - Gray: Switch is inactive / not working.
7, 8	Limit switch 3-4	- Displays the status of the Cooked Zone limit switch. - Green: Switch is active. - Gray: Switch is inactive.
9	Emergency stop 1, 2	- Monitors the status of the Raw Zone Emergency button and the internal Cabinet Emergency button. - A flashing red indicator appears if either Emergency button is pressed.
10	Emergency stop 3	- Monitors the status of the Cooked Zone Emergency button. - A flashing red indicator appears if the Emergency button is pressed.
11	Door 1 lock status	- Monitors the locking mechanism of the Raw Zone door. - Green: Door is securely locked. - Gray: Door is unlocked.
12	Door 2 lock status	- Monitors the locking mechanism of the Cooked Zone door. - Green: Door is securely locked. - Gray: Door is unlocked.

HMI control panel		
13	Display time	Displays the remaining countdown time for the current sterilization cycle while the UV lamps are active.
14-25	UV lamp status	- Monitors the operational status and health of individual UV lamps. - All Green (During Operation): System is running normally; all lamps are functioning. - All Red (Stopped): System is inactive; all lamps are off. - Specific Red (During Operation): Indicates a malfunction or failure of that specific UV lamp.

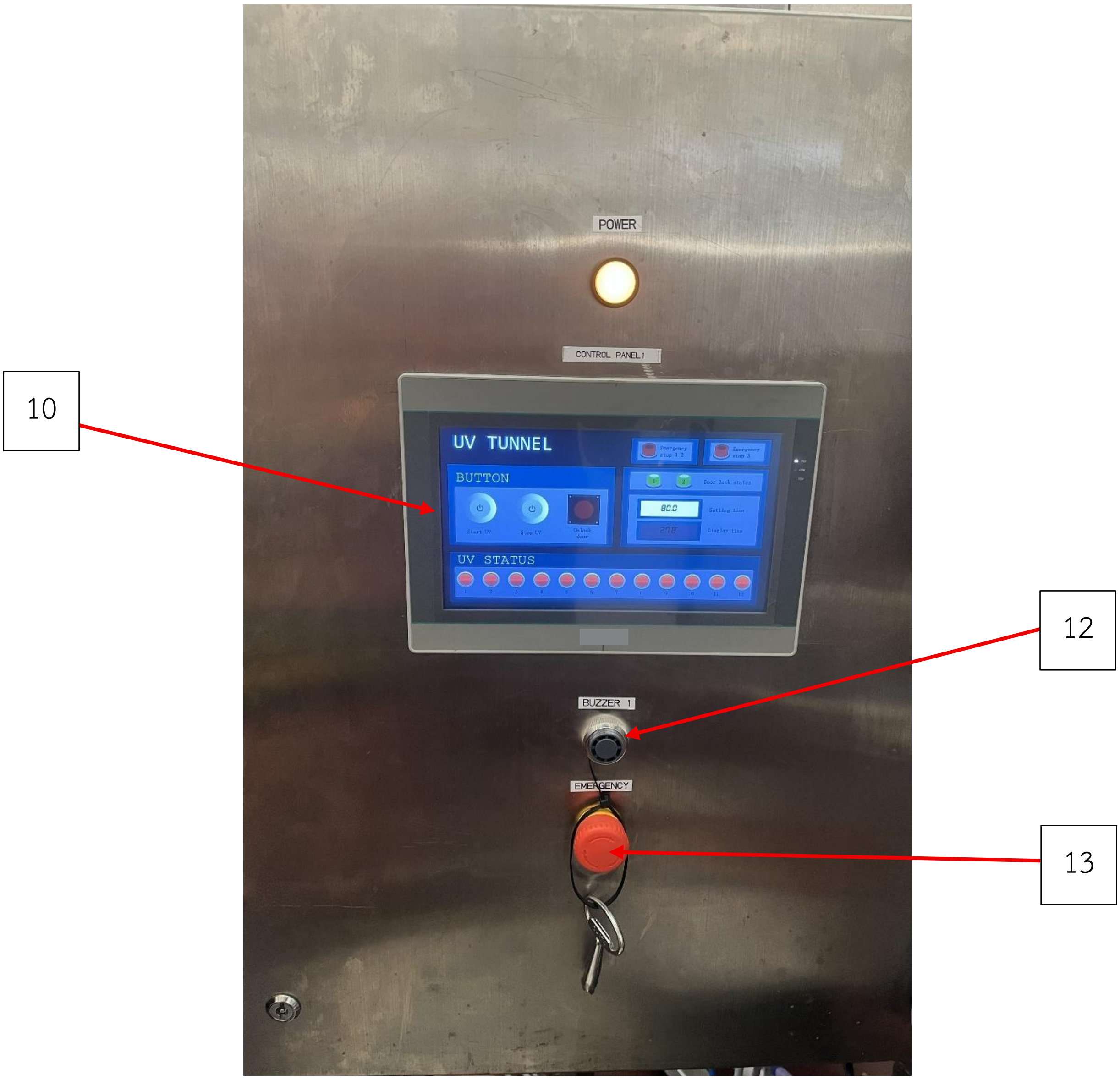
6. Electrical Control Panel

6.1 Electrical Control Panel Raw Zone

Inside

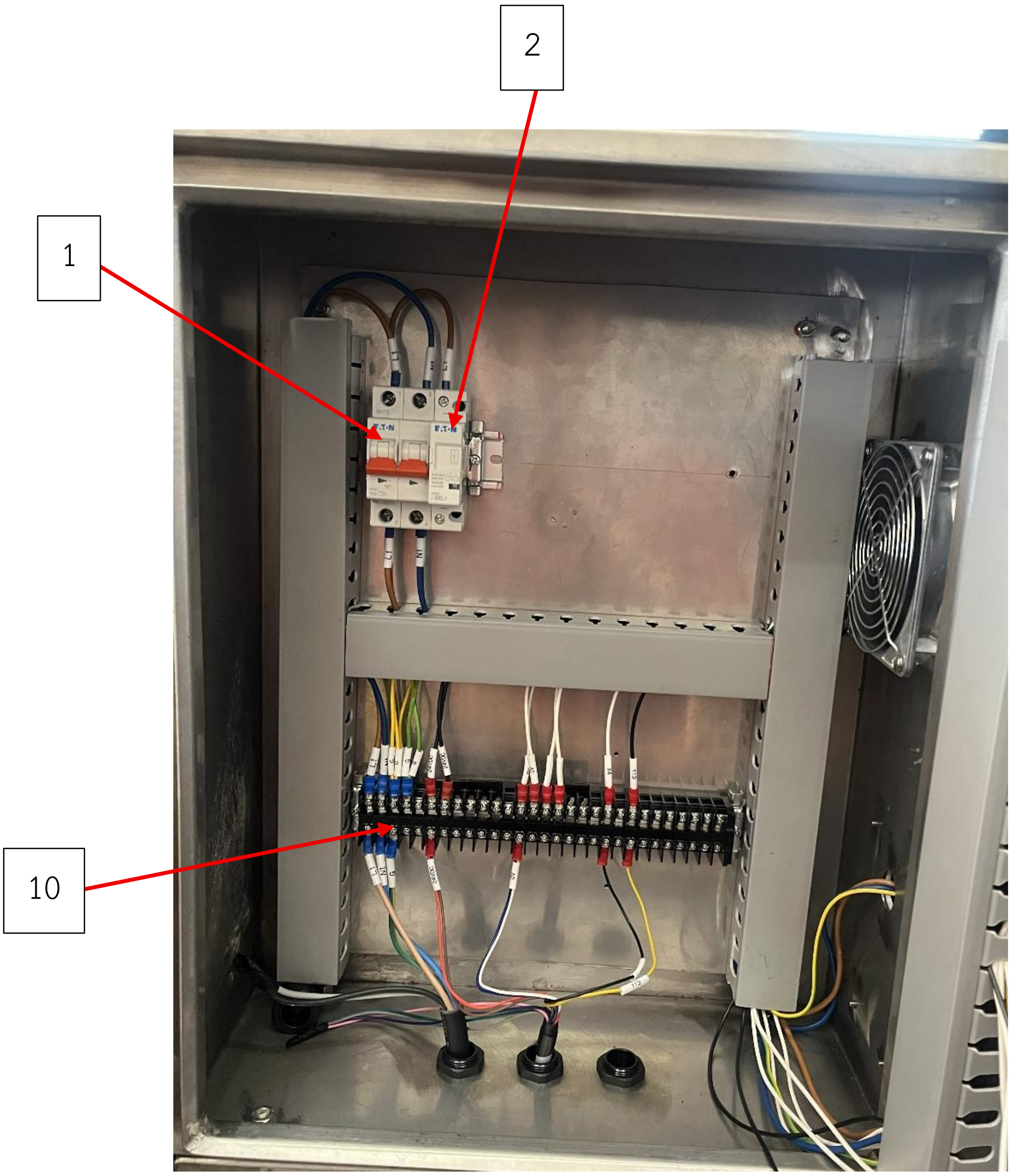


Outside

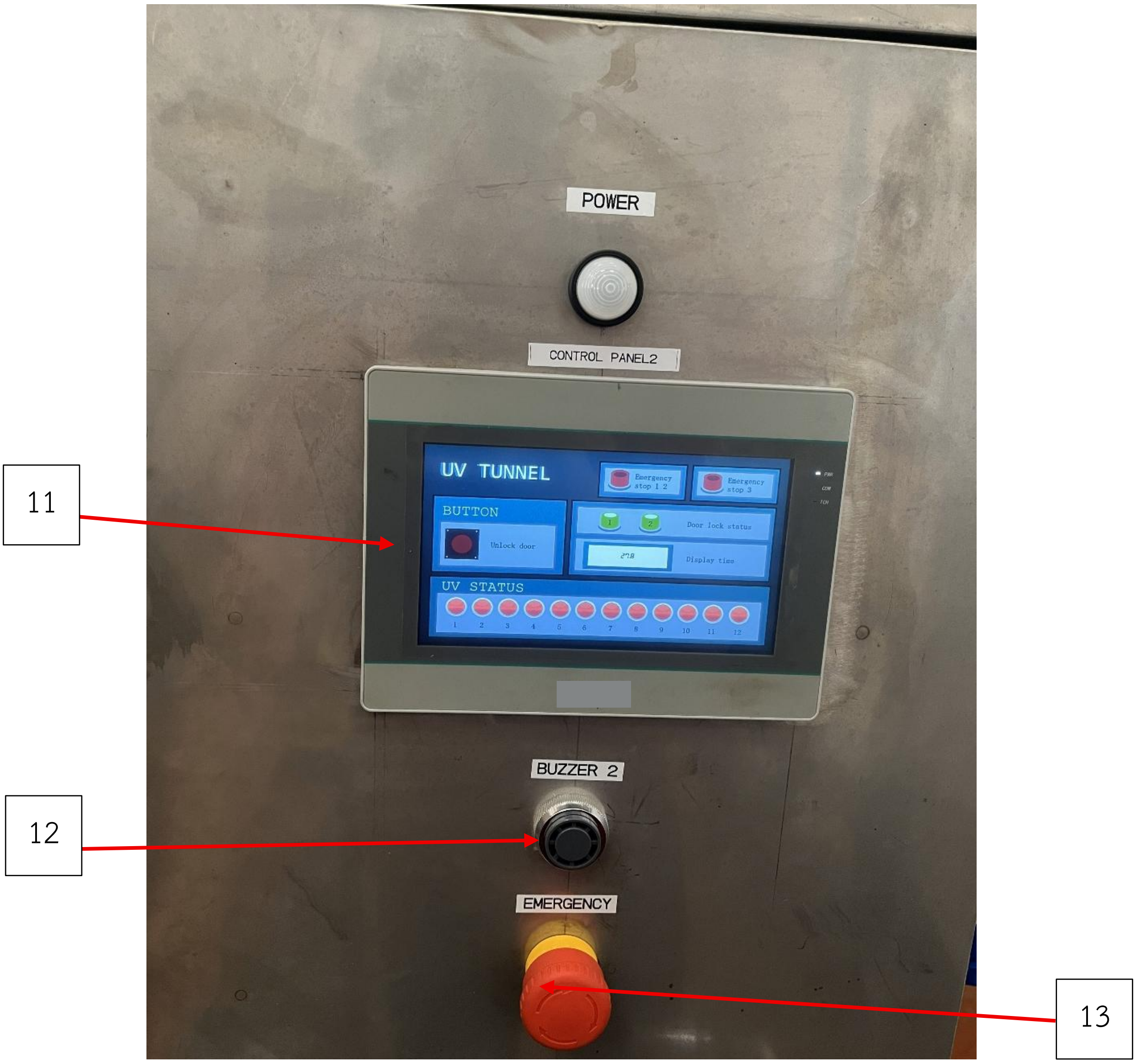

10
12
13
POWER
CONTROL PANEL 1
UV TUNNEL
BUTTON
UV STATUS
BUZZER 1
EMERGENCY

6.2 Electrical Control Panel Cooked Zone

Inside



Outside


POWER
CONTROL PANEL2
UV TUNNEL
BUTTON
Unlock door
Door lock status
Display time
UV STATUS
BUZZER 2
EMERGENCY
11
12
13

6.3 List of Electrical Control Panel spare part

Controller box part list		
Item	Device Name	Oty
1	Breaker	1
2	Fuse 8 A	3
3	Power supply 5 A	1
4	PLC	1
5	Switching hub	1
6	Relay 24 VDC	6
7	Magnetic	6
8	Motor circuit breaker 2.5 A	12
9	Slim relay 24 VDC	12
10	HMI	1
11	HMI	1
12	Buzzer	2
13	Emergency button	3

7. Operating Instructions

1. Press the "Unlock door" button on the HMI to open the door. The "Door lock status" indicator will change from Green to Gray.
2. Push the food cart into the UV tunnel and ensure the door is closed tightly. Once secured, the "Door lock status" will display Green for both sides, confirming that the system is locked.
3. Case: Transferring carts from Raw Zone to Cooked Zone

3.1 Verify that all "Door lock status" and "Limit switch" indicators are Green. If confirmed, press "Start UV" to begin sterilization.

3.2 Upon completion of the cycle, a Buzzer will sound. Open the door on the Cooked Zone side to remove the food cart.

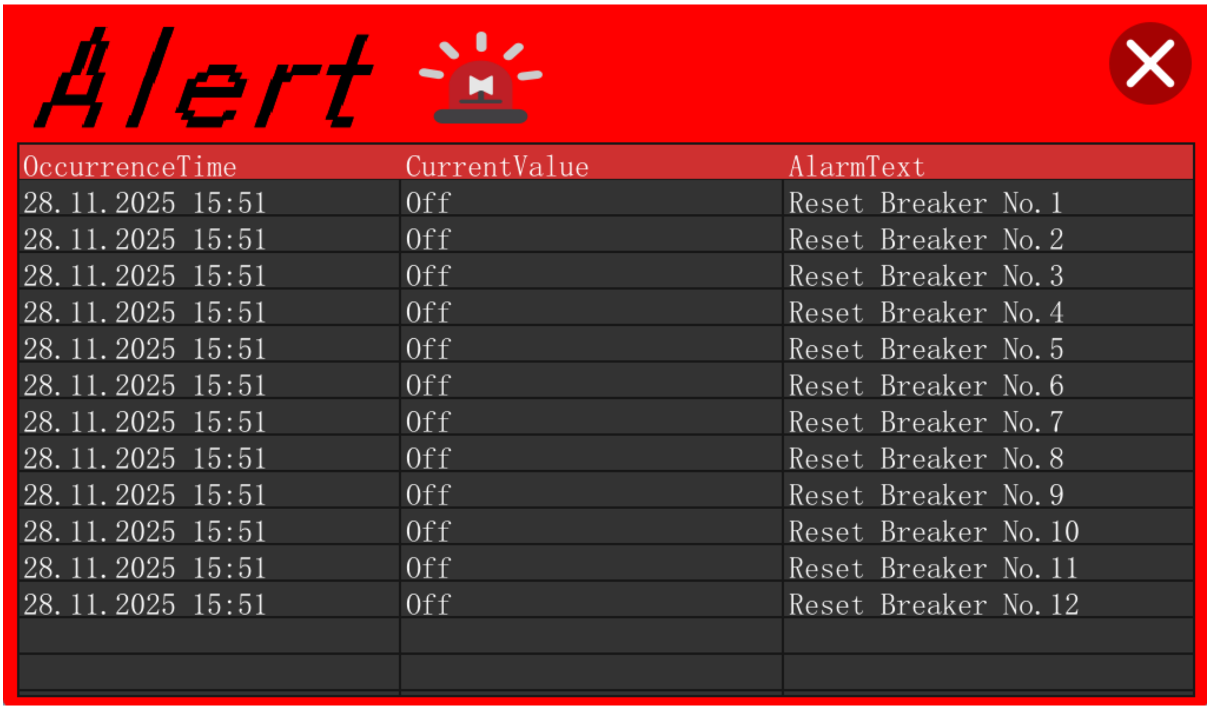
4. Case: Transferring carts from Cooked Zone to Raw Zone

4.1 Verify that all "Door lock status" and "Limit switch" indicators are Green. If any indicator is not Green, open and close the door again to reset the limit switches.

4.2 Carts returning from the Cooked Zone are empty; therefore, sterilization is not required.

5. The "Unlock door" button is disabled and cannot be used while the UV lamps are active.
6. There are 3 Emergency buttons located at: Raw Zone electrical control panel, the Cooked Zone electrical control panel, and inside the UV Tunnel. Pressing the internal Emergency button will unlock the Raw Zone door.
7. If an Emergency button is pressed while UV lamps are active, the system will immediately stop the UV operation. The door on the corresponding side will unlock, and the HMI will display a flashing red light indicating the location of the emergency.
8. If an Emergency button is already active in one zone, pressing the Emergency button in the other zone will have no effect (e.g., if the Raw Zone Emergency is active, the Cooked Zone Emergency is disabled).
9. To Reset: Release the Emergency button, then open and close the door to reset the limit switch. The Emergency status light will turn off, and the tunnel will return to normal operation.

10. Safety Interlock: If any limit switch status changes (light turns off/not Green) during UV operation, the system will immediately cut power to stop the UV process.
11. UV Lamp Failure & Protection: In the event of a UV lamp failure (short circuit or damage), the Motor Circuit Breaker will trip immediately to cut power for safety. The HMI screen will display an "Alert" window notifying the operator that a Motor Circuit Breaker Trip has occurred.



OccurrenceTime	CurrentValue	AlarmText
28.11.2025 15:51	Off	Reset Breaker No.1
28.11.2025 15:51	Off	Reset Breaker No.2
28.11.2025 15:51	Off	Reset Breaker No.3
28.11.2025 15:51	Off	Reset Breaker No.4
28.11.2025 15:51	Off	Reset Breaker No.5
28.11.2025 15:51	Off	Reset Breaker No.6
28.11.2025 15:51	Off	Reset Breaker No.7
28.11.2025 15:51	Off	Reset Breaker No.8
28.11.2025 15:51	Off	Reset Breaker No.9
28.11.2025 15:51	Off	Reset Breaker No.10
28.11.2025 15:51	Off	Reset Breaker No.11
28.11.2025 15:51	Off	Reset Breaker No.12

Example of the alert screen when a UV lamp gets damaged.

8. Program Specifications

In the event of future program modifications, this Label Table provides a detailed explanation of the functions associated with each Address. The "Source" column categorizes the type of each Address as follows :

1. **Out source**: An Address used to directly command external devices or receive input commands from devices.
2. **Function**: An Address serving as internal logic or conditions within the program; it is not directly hardwired to external devices.
3. **HMI**: An Address utilized on the HMI screen to display operational status or functioning as a control button to execute programmed commands.

Label			
Address	Device	Description	Source
X0	Start UV	Activate the UV lamp operation.	HMI
X1	Stop UV	Disable the UV lamp operation.	HMI
X2	Emer low	Emergency Stop (Raw Zone): Cuts power to all UV lamps and unlocks the Raw Zone door.	Out source
X3	Emer high	Emergency Stop (Cooked Zone): Cuts power to all UV lamps and unlocks the Cooked Zone door.	Out source
X4	Unlock door low	Commands the Raw Zone Cylinder to unlock the door.	HMI
X5	Unlock door high	Commands the Cooked Zone Cylinder to unlock the door.	HMI
X10-13	Limit switch 1-4	Monitors the status of limit switches (Raw/Cooked) to verify consistency with Cylinder operation.	Out source
X20-33	Motor Circuit breaker 1-12	Trips/Cuts power to UV lamps in the event of a lamp failure/damage.	Out source
Y0	Cylinder 1	Controls the operation of the Raw Zone Cylinder.	Out source

Y1	Cylinder 2	Controls the operation of the Cooked Zone Cylinder.	Out source
Y2-7	Relay 1-6	Relays used to switch UV lamps On/Off.	Out source
Y10	Buzzer	Activates the Buzzer signal.	Out source
M0	Door 1 lock	Controlled by inputs X10, X11 to lock the Raw Zone door (Y0).	HMI
M1	Set UV ON	Activates when receiving commands from Y2-7 and T0 to turn ON the UV lamps.	Function
M2	Door 1 unlock	Receives commands from X2, X4 to Reset Y0 (Unlock the Raw Zone door).	Function
M4	Emer Door 1 Command	Activates upon receiving a signal from the HMI to cancel the Emergency status.	HMI
M5	Door 2 unlock	Controlled by inputs X12, X13 to lock the Cooked Zone door (Y1).	HMI
M6	Door 2 unlock	Receives commands from X3, X5 to Reset Y1 (Unlock the Cooked Zone door).	Function
M7	Emer Door 2 Command	When ON, triggers a flashing indicator on the HMI to signal that the Cooked Zone Emergency button is pressed.	Function
M9	Door 2 lock	Interlock: When active, prevents the Cooked Zone door from being unlocked.	Function
M10	Door 1 lock	Interlock: When active, prevents the Raw Zone Area door from being unlocked.	Function
M29	Unlock door 1 command	Processes commands from X1 combined with condition M29 or X4 to trigger the unlocking of the Raw Zone door.	Out source
M77	Alert	Triggers the "Alert" window on the HMI in the event of a Circuit Breaker trip or UV lamp failure.	HMI

M99	UV Set off	When ON, turns off the UV lamp status indicators on the HMI screen.	Function
M100-111	UV 1-12	Displays the operational status of each individual UV lamp on the HMI.	HMI
T0	Timer	Controls the duration of the UV sterilization cycle.	Function
D0	Input Number	Input field for setting the desired UV operation time (Duration).	HMI
D1	Display Number	Displays the remaining countdown time of the UV cycle on the HMI screen.	HMI